Smart Chair

Smart Chair Application —

1. Project Requirements

The Smart Chair application consists of two parts:

- **SmartChairBackend** — Python Flask backend, MongoDB Atlas, JWT authentication, and ML model
- **SmartChairApp** — React Native mobile application built with Expo

You will also need an internet connection and an Android/iOS phone with **Expo Go** installed.

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2. Software to Install

Install the following software before setting up the project:

Software	Version	Purpose	Download
Node.js	LTS 20+	Runs the React Native/Expo application	https://nodejs.org
Python	3.10+	Runs the Flask backend and ML model	https://python.org
Git	Latest	Source code/version control	https://git-scm.com
VS Code	Latest	Recommended code editor	https://code.visualstudio.com
Expo Go	Latest	Runs the mobile application on your phone	Play Store / App Store

Recommended

After installing Node.js and Python, verify the installations:

node --version

npm --version

python --version

On some systems, you may need:

python3 --version

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3. Project Folder Structure

The complete project should look like this:

```
SmartChair/
|
├── SmartChairBackend/
|   ├── app.py
|   ├── model.pkl
|   ├── requirements.txt
|   ├── import_data.py
|   ├── chair_data.csv
|   └── .env
|
└── SmartChairApp/
    ├── App.js
    ├── app.json
    ├── package.json
    ├── index.js
   |
   └── src/
       ├── screens/
       |   ├── LoginScreen.js
       |   ├── CalendarScreen.js
       |   └── AnalyticsScreen.js
       |
       ├── components/
       |   └── PostureAnimation.js
       |
       └── api/
           └── apiService.js
```

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4. Backend Dependencies

The backend uses Python and Flask.

The preferred method is to provide the requirements.txt file and run:

pip install -r requirements.txt

If your system uses pip3:

pip3 install -r requirements.txt

Backend packages

Package	Version	Purpose
flask	3.0.0	Backend web framework
flask-jwt-extended	4.6.0	JWT authentication
pymongo	4.6.0	MongoDB Atlas connection
bcrypt	4.1.2	Password hashing
scikit-learn	1.4.0	ML model inference
python-dotenv	1.0.0	Loads environment variables from .env

To install them manually:

pip install flask==3.0.0 flask-jwt-extended==4.6.0 pymongo==4.6.0 bcrypt==4.1.2 scikit-learn==1.4.0 python-dotenv==1.0.0

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5. Frontend Dependencies

The mobile application is built using **React Native + Expo**.

Go inside the SmartChairApp folder and run:

npm install

This automatically installs all dependencies specified in package.json.

Main frontend packages

Package	Purpose
expo	Core Expo SDK
react-native	Mobile application framework
@react-navigation/native	Screen navigation

Package	Purpose
@react-navigation/stack	Stack-based navigation
react-native-screens	Native screen components
react-native-safe-area-context	Safe area handling
react-native-gesture-handler	Touch and gesture support
react-native-reanimated	Animations
react-native-calendars	Calendar component
axios	API/HTTP requests
@react-native-async-storage/async-storage	Local token storage
react-native-animatable	Posture animations
expo-linear-gradient	Gradient UI backgrounds

Three Critical Items

1. model.pkl

This is the trained machine-learning model.

It **must** be present inside:

SmartChairBackend/

Without model.pkl, the backend may fail when loading the ML model.

2. .env

This contains the environment/configuration values required by the backend, including the MongoDB connection information.

It **must** be present inside:

SmartChairBackend/

Without the required .env configuration, the backend will not be able to connect to MongoDB.

3. Laptop IP Address

The mobile application communicates with the backend through the laptop's local IP address.

The IP address in:

SmartChairApp/src/api/apiService.js

must be changed to the IP address of the laptop running the backend.

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6. Complete Installation Steps

Step 1 — Copy the Project

Copy the complete SmartChair folder to the laptop.

For example:

Documents/

└── SmartChair/

Step 2 — Install Required Software

Install:

1. Node.js LTS 20+
2. Python 3.10+
3. Git
4. VS Code
5. Expo Go on the mobile phone

Step 3 — Install Backend Dependencies

Open a terminal inside:

SmartChairBackend

Run:

pip install -r requirements.txt

or:

pip3 install -r requirements.txt

Step 4 — Install Frontend Dependencies

Open another terminal inside:

SmartChairApp

Run:

npm install

Wait until all packages are installed successfully.

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8. Configure the Laptop IP Address

The phone and laptop need to be connected to the **same Wi-Fi network**.

Find the laptop's local IP address.

Windows

Open Command Prompt and run:

ipconfig

Look for:

IPv4 Address

Example:

192.168.1.10

macOS

Run:

ifconfig | grep "inet " | grep -v 127.0.0.1

Step 9 — Update apiService.js

Open:

SmartChairApp/src/api/apiService.js

Find the BASE_URL configuration.

Replace the existing IP address with the laptop's current IP address.

For example:

http://192.168.1.10:5000

Use the **actual IP address of the laptop**.

Do not use:

localhost

or:

127.0.0.1

for the mobile app, because on the phone those addresses refer to the phone itself rather than the laptop.

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10. Start the Backend

Open **Terminal 1**.

Go to the backend folder:

cd SmartChairBackend

Start the Flask server:

python app.py

If your system requires python3:

python3 app.py

Keep this terminal running.

The backend must remain running while using the mobile application.

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11. Start the Mobile Application

Open **Terminal 2**.

Go to the frontend folder:

cd SmartChairApp

Start Expo:

npx expo start --clear

Expo will display a QR code in the terminal/browser.

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12. Open the App on the Phone

1. Install **Expo Go** on the phone.
2. Connect the phone to the **same Wi-Fi network as the laptop**.
3. Open Expo Go.
4. Scan the QR code generated by Expo.
5. The Smart Chair application should open.

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13. If the App Cannot Connect to the Backend

Check the following:

Laptop and phone

Make sure both are connected to the **same Wi-Fi network**.

Backend

Make sure:

python app.py

is still running.

IP address

Check:

SmartChairApp/src/api/apiService.js

and make sure the IP address matches the laptop's current IPv4 address.

Firewall

If the backend is running but the phone cannot connect, the laptop's firewall may be blocking the Flask port.

.env

Make sure:

SmartChairBackend/.env

exists and contains the required configuration.

model.pkl

Make sure:

SmartChairBackend/model.pkl

exists.

Final Startup Commands

Once everything is configured, you only need two terminals.

Terminal 1 — Backend

cd SmartChairBackend

python app.py

Terminal 2 — Mobile App

cd SmartChairApp

npx expo start --clear

Then open **Expo Go** on the phone and scan the QR code.